Whole-Mount Indirect Fluorescent Immunohistochemistry – Double or multiple staining

Gaido Group, GUDMAP Consortium

*All incubations and washes are carried out at 4 ºC on an orbital shaker.

1. Tissue samples are fixed with 4% paraformaldehyde in PBS (1:4, purchased 16%, diluted with PBS), overnight, 4 ºC. (Keep samples in tubes?)
2. Rinse, 2 times in PBS.
3. Quench in 50 mM NH4CL, 15 min.
4. Block in blocking buffer, 24 hr (1% BSA, 0.1% Saponin, 0.02% sodium azide in PBS) [100mg BSA+10ul Saponin+2ul sodium azide ------à10 ml PBS]. You also need to try blocking with Triton X-100 (0.1%) instead of saponin.
5. Incubated in FIRST primary antibody in fresh blocking buffer, 24 hr, 4 ºC. You can incubate all primary antibodies at the same time.
6. Wash 5 times, 1 hr/each, in blocking buffer.
7. Incubated in secondary antibody in fresh blocking buffer, 24 hr, 4 ºC. You can incubate all secondary antibodies at the same time.
8. Wash 5 times, 1 hr/each, in blocking buffer.

Fluorescently labeled samples are image by laser scanning confocal microscope. Whole-mount samples are viewed in antifade buffer with glycerol in a culture dish with attached coverslip. For viewing, use the special coverslip glass-bottomed culture dishes without a coverslip.